

Guilherme Ribeiro Figueira

Junior C++ Software Engineer

📍 Lisbon, Portugal • ✉ gr.figueira@proton.me • 🌐 grfigueira.github.io • 🔗 linkedin.com/in/grfigueira • 🐙 github.com/grfigueira

Experience

C++ Software Engineer

Critical Techworks | BMW Group

Lisbon, Portugal (Hybrid)

Jan 2026 – present

- Developed safety-critical C++17 real-time software for an ECU on **unannounced BMW vehicles**.
- **Embedded programming using RTOS** on ARM processors. Strict **memory safety** and **MISRA** compliant programming
- Contributed to multiple embedded platform applications such as check control messages, car maintenance monitoring, customer personalization of vehicle features and management of vehicle DTCs.
- Devised and implemented a **data-driven architecture** for an internal vehicle state management component.
- Contributed to the development of the **middleware layer of a central ECU** with bit-packed PDU communication over message queues, SOME/IP and CAN.
- Maintained high test coverage via **TDD** and wrote easy to read **documentation**. Followed **Agile** and **Scrum** methodologies.

Unity XR Developer - Curricular Internship

Glartek

Leiria, Portugal (Hybrid)

Mar 2023 – Aug 2023

- Built an **AR 3D asset inspection tool** for a **Microsoft HoloLens 2** industry 4.0 project.
- Collaborated with the web team to design a pipeline for dynamic loading of 3D assets from backoffice into AR.
- Used linear algebra, graph search algorithms and caching to reliably process any 3D model with **optimal performance**.

Projects

Papoila Game Engine | Game Engine | (C++17, ImGui, GLFW)

WIP C++17 **OpenGL game engine** with assimp model loading, textured mesh rendering, cubemap skyboxes, Phong lighting, editor camera controls, and proximity dithering built in.

Easy Dialog Editor | Tool | (C++17, ImGui, SDL)

Lightweight **visual node editor** for **non-linear dialogue trees** in games. Exports dialogue data to JSON for engine agnostic integration. The entire program is a portable 2.7 MB executable with no external dependencies.

The Natural's Descent | Game | (Unity, C#)

Implemented procedural cave generation with **Wave Function Collapse**, AI combat mechanics, and a data-oriented grid puzzle system for easy design. Worked in a team of 2.

The Shape Of Silence | Game | (UE5, C++, Blueprint, FMOD)

Narrative puzzle game with C++ systems for interaction, inventory, item inspection, and **data-driven** puzzles via **Unreal Data Assets**, exposing designer-friendly **Blueprint** delegates. Non-linear dialogue authored with my own **EasyDialogueEditor** and **FMOD** audio.

More projects, further details, videos and source code can be found in my [Portfolio](#) and [Github](#).

Skills

Programming Languages: C++, C#, Python, LUA, Java, TypeScript, OCaml

Graphics: OpenGL, Vulkan, ImGui, GLFW, SDL3, WebGL, OpenCV, QGIS

Engines & Tools: Unreal Engine 5, Unity, FMOD, Blender, Photoshop, Audacity

Build & Documentation: Bazel, CMAKE, Doxygen, Sphinx, Jinja

Workflow tools: RenderDoc, Docker, Linux Shell, VIM, Git, Jira

Databases: PostgreSQL, SQLite

Languages: Portuguese (native), English (proficient)

Education

Nova School of Science and Technology

B.Sc. + M.Sc. in Computer Science and Engineering

Lisbon, Portugal

Sept 2020 – Jan 2025

- Master Thesis (**Grade: 19/20**): Developed a Unity MR authoring tool that dynamically transforms scanned rooms into tangible virtual environments using Quest 3 scene understanding, procedural placement rules, and tracked-object substitution.
- Implemented performant stencil-buffer portals for MR transitions, stress-tested with ~200 simultaneous portals on Quest 3 hardware running at a stable framerate.

Publications

Towards Augmented XR Movement and Interaction Confidence across Substitutional Reality

Jan 2025

6th place (out of 25) on public vote for best presentation

Guilherme R. Figueira, Rui Nóbrega

[10.1109/ICGI68463.2025.11302724](#) (International Conference on Graphics and Interaction (ICGI), 2025)